

		$x = 1.2 \times 10^{-3} \text{ m} = 0.12 \text{ mm}$
18	B	The electric field lines must always be perpendicular to the surfaces of the charged conductors (i.e. the sphere and the metal plate). The electric field strength is strongest in the region of closest separation. Option A: The central region of the metal plate appears to have a weaker electric field strength than at the sides. Field lines are not normal to the surface. Option C: The electric field strength seems to be evenly spread out along the metal plate. Field lines are not normal to the surface.

		Option D: The electric field strength seems to be confined to just one section of the metal plate.
19	B	The electric fields at the middle point due to X and Y both act towards the right. Thus, the net electric field strength is equal to the sum of the two individual electric fields.